Electronique 1

Convertisseurs DC-DC

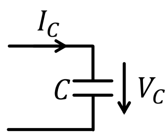
DC/DC Wandler

Condensateurs et Inductances
Kondensatoren und Induktivitäten

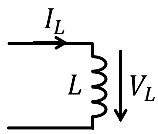
dérivé et intégral

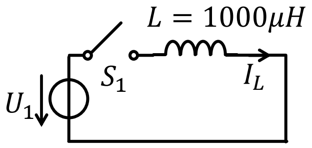
Derivat und Integral

$$I_C(t) = C\frac{dV_C(t)}{dt} \qquad V_C(t) = \frac{1}{C}\int_{t_0}^{t} I_C(\tau)d\tau + V_C(t_0)$$



$$V_L(t) = L\frac{dI_L(t)}{dt} \qquad I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$



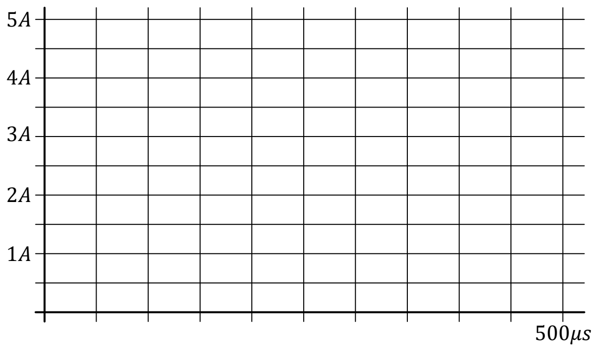


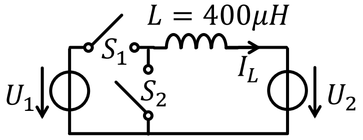
Exercice: À temps $t = 0$ l'interrupteur S_1 est fermé. Dessiné le courent $I_L(t)$ jusqu'à $t = 500\mu s$.

Aufgabe: Zum Zeitpunkt $t = 0$ ist der Schalter S_1 geschlossen. Zeichne den Lauf $I_L(t)$ bis $t = 500\mu s$.

$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

$U_1 = 5V \qquad I_L(0) = 1\,A$





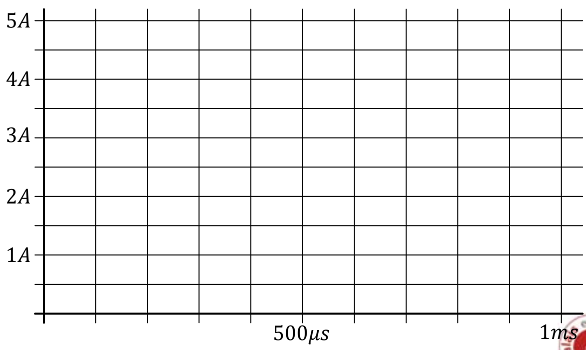
Exercice: À temps $t = 0$ l'interrupteur S_1 est fermé. Dessiné le courent $I_L(t)$ jusqu'à $t = 400\mu s$. À $t = 400\mu s$ l'interrupteur S_1 est ouvert et l'interrupteur S_2 est fermé. Dessiné le courent $I_L(t)$ jusqu'à il attente son valeur initial. Ça prendre combien de temps? Quel est la valeur moyenne de courent I_L?

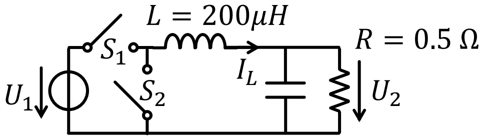
$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

$U_1 = 5V \qquad I_L(0) = 1\,A$

$U_2 = 2V$

Aufgabe: Zum Zeitpunkt $t = 0$ ist der Schalter S_1 geschlossen. Zeichne den Lauf $I_L(t)$ bis $t = 400\mu s$. Bei $t = 400\mu s$ ist Schalter S_1 geöffnet und Schalter S_2 geschlossen. Zeichnen Sie den Lauf $I_L(t)$, bis er seinen Anfangswert erreicht. Wie lange wird es dauern? Was ist der Durchschnittswert des Laufens I_L?



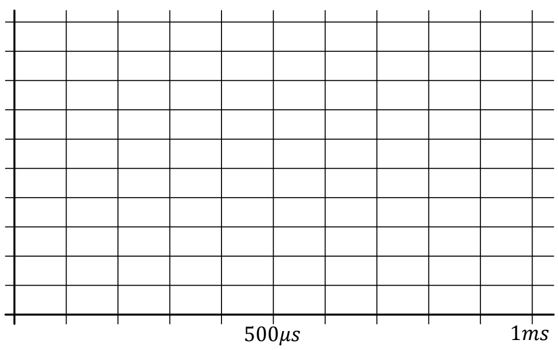


$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

$U_1 = 5V$

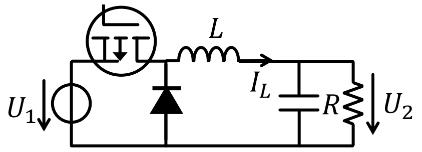
Exercice: À temps $t = 0$ l'interrupteur S_1 est fermé et l'interrupteur S_2 est ouvert. À $t = 300\mu s$ l'interrupteur S_1 est ouvert et l'interrupteur S_2 est fermé. À $t = 1\ ms$ le courent attente son valeur initial. Le condensateur est suffisamment grand pour que la tension U_2 soit constante. Quel est la tension U_2? Pour le de courent I_L, quel sont les valeurs d'ondulation (ΔI_L), pic $(\hat{I}_L)$, moyenne $(\bar{I}_L)$ et initial/minimum (I_{L0})?

Aufgabe: Zum Zeitpunkt $t = 0$ ist Schalter S_1 geschlossen und Schalter S_2 geöffnet. Bei $t = 300\mu s$ ist Schalter S_1 geöffnet und Schalter S_2 geschlossen. Bei $t = 1\ ms$ erwartet der Strom seinen Anfangswert. Der Kondensator ist groß genug, dass die Spannung U_2 konstant ist. Wie hoch ist die Spannung U_2 ? Was sind für den Lauf I_L die Werte von Welligkeit (ΔI_L) , Spitzenwerte $(\hat{I}_L)$ Durchschnitt $(\bar{I}_L)$ und Anfang/Minimum (I_{L0})?



- Zum Zeitpunkt $t = 0$ ist Schalter S_1 geschlossen und Schalter S_2 geöffnet.
- Bei $t = T_1$ ist Schalter S_1 geöffnet und Schalter S_2 geschlossen.
- Bei $t = T$ ist Schalter S_1 geschlossen und Schalter S_2 geöffnet.
- Und so weiter

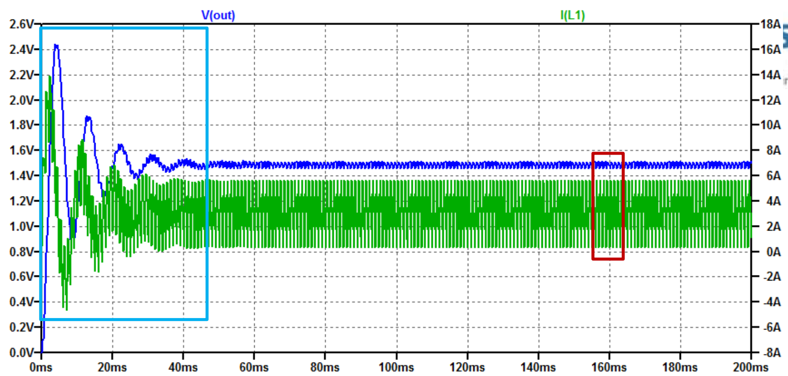
- À temps $t = 0$ l'interrupteur S_1 est fermé et l'interrupteur S_2 est ouvert.
- À $t = T_1$ l'interrupteur S_1 est ouvert et l'interrupteur S_2 est fermé.
- À $t = T$ l'interrupteur S_1 est fermé et l'interrupteur S_2 est ouvert.
- Et ainsi de suite



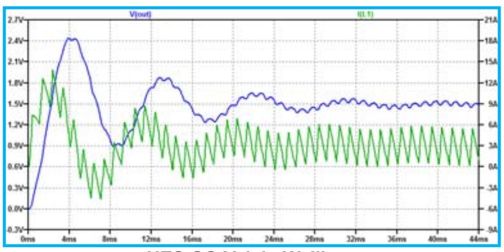
PMOS - FDS4953
Diode - 1N914
C=4.7 mF

- Schalter S_1 wird oft mit einem MOSFET implementiert
- Der Schalter S_2 kann mit einem MOSFET oder einer Diode implementiert werden.
- $I_L \geq 0$, $U_2 \leq U_1$, $U_2 = U_1 \frac{T_1}{T}$
- Die Baugruppe wird BUCK-Konverter genannt

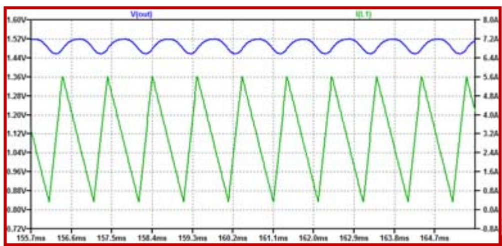
- L'interrupteur S_1 est souvent implémenté avec un MOSFET
- L'interrupteur S_2 est peut être implémenté avec un MOSFET ou une diode.
- $I_L \geq 0$, $U_2 \leq U_1$, $U_2 = U_1 \frac{T_1}{T}$
- Le montage s'appel le convertisseur BUCK



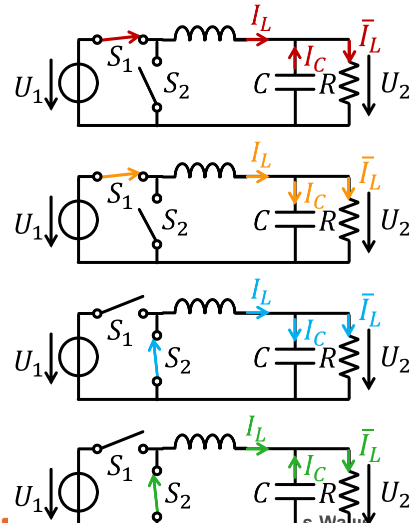
Régime transitoire Ausgleichsvorgang

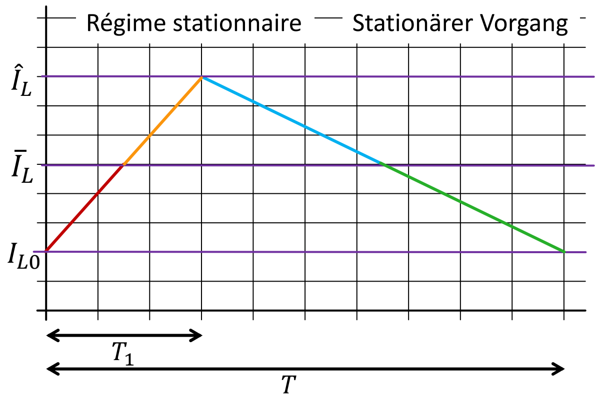


Régime stationnaire Stationärer Vorgang



Condensateur et ondulation de tension
Kondensator und Spannungswelligkeit

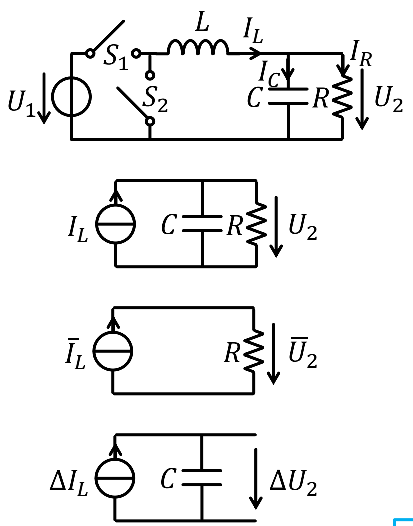


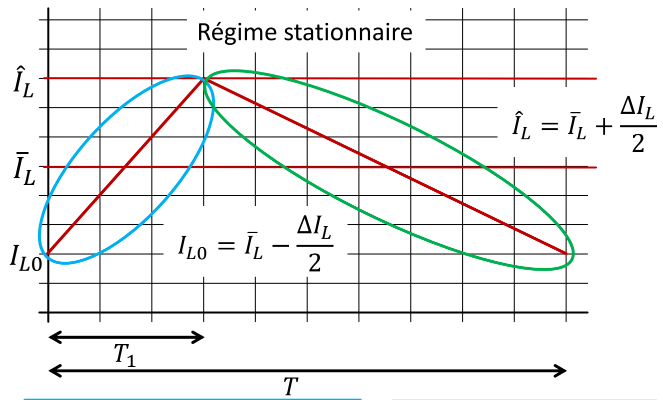


$I_R = \bar{I}_L$

$I_C = I_L(t) - \bar{I}_L = \Delta I_L(t)$

Condensateur et ondulation de tension
Kondensator und Spannungswelligkeit



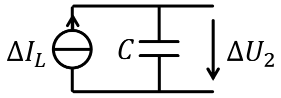


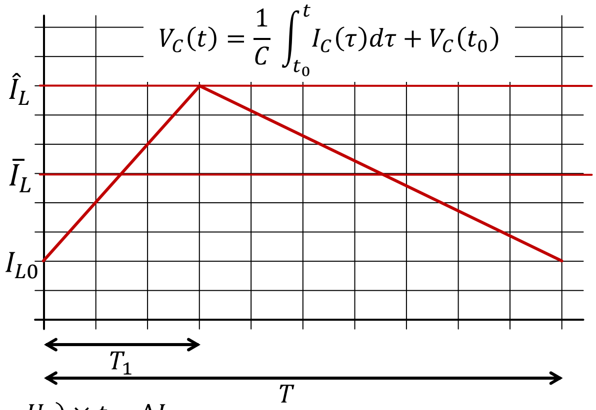
$$\boxed{I_L(t) = \frac{(U_1 - U_2) \times t}{L} + I_{L0}} \qquad \boxed{I_L(t) = -\frac{U_2 \times t}{L} + \hat{I}_L}$$

$$\Delta I_L(t) = I_L(t) - \bar{I}_L$$

$$\boxed{\Delta I_L(t) = \frac{(U_1 - U_2) \times t}{L} - \frac{\Delta I_L}{2}} \qquad \boxed{\Delta I_L(t) = -\frac{U_2 \times t}{L} + \frac{\Delta I_L}{2}}$$

Condensateur et ondulation de tension
Kondensator und Spannungswelligkeit



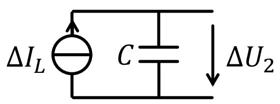


$$\Delta I_L(t) = \frac{(U_1 - U_2) \times t}{L} - \frac{\Delta I_L}{2} \qquad \Delta I_L(t) = -\frac{U_2 \times t}{L} + \frac{\Delta I_L}{2}$$

$$V_C(t) = \frac{(U_1 - U_2) \times t^2}{2LC} - \frac{\Delta I_L}{2C} t + V_{C0} \qquad V_C(t) = -\frac{U_2 \times t^2}{2LC} + \frac{\Delta I_L}{2C} t + V_{C1}$$

$$\Delta V_C = V_{C,max} - V_{C,min}$$

Condensateur et ondulation de tension
Kondensator und Spannungswelligkeit



$$\Delta I_L(t) = \frac{(U_1 - U_2) \times t}{L} - \frac{\Delta I_L}{2}$$

$$V_C(t) = \frac{(U_1 - U_2) \times t^2}{2LC} - \frac{\Delta I_L}{2C} t + V_{C0}$$

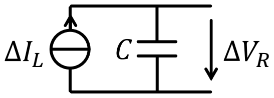
V_C maximal ou minimal quand $\frac{dV_C}{dt} = \Delta I_L(t) = 0$

$$\Delta I_L(t) = \frac{(U_1 - U_2) \times t}{L} - \frac{\Delta I_L}{2} = 0 \rightarrow t = \frac{\Delta I_L \times L}{2(U_1 - U_2)}$$

$$V_{C,max,min} = \frac{(U_1 - U_2)}{2LC} \frac{{\Delta I_L}^2 \times L^2}{4(U_1 - U_2)^2} - \frac{\Delta I_L}{4C} \frac{\Delta I_L \times L}{(U_1 - U_2)} + V_{C0}$$

$$V_{C,min} = \frac{{\Delta I_L}^2 \times L}{8C(U_1 - U_2)} - \frac{{\Delta I_L}^2 \times L}{4C(U_1 - U_2)} + V_{C0} = -\frac{{\Delta I_L}^2 \times L}{8C(U_1 - U_2)} - V_{C0}$$

Condensateur et ondulation de tension
Kondensator und Spannungswelligkeit



$$\Delta I_L(t) = -\frac{U_2 \times t}{L} + \frac{\Delta I_L}{2}$$

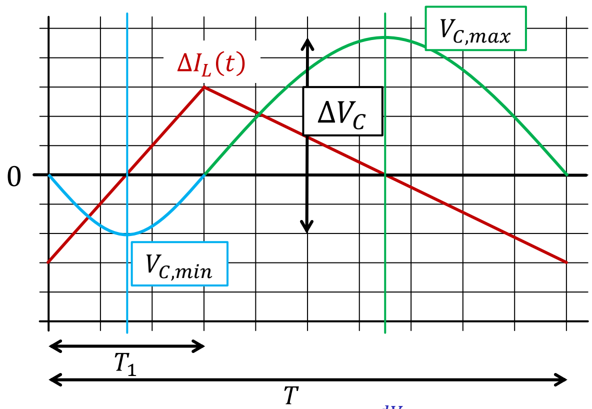
$$V_C(t) = -\frac{U_2 \times t^2}{2LC} + \frac{\Delta I_L}{2C} t + V_{C1}$$

V_C maximal ou minimal quand $\frac{dV_C}{dt} = \Delta I_L(t) = 0$

$$\Delta I_L(t) = -\frac{U_2 \times t}{L} + \frac{\Delta I_L}{2} = 0 \rightarrow t = \frac{\Delta I_L \times L}{2U_2}$$

$$V_{C,max} = -\frac{U_2}{2LC}\left(\frac{\Delta I_L \times L}{2U_2}\right)^2 + \frac{\Delta I_L}{2C}\left(\frac{\Delta I_L \times L}{2U_2}\right) + V_{C1}$$

$$V_{C,max} = -\frac{\Delta I_L^2 \times L}{8CU_2} + \frac{\Delta I_L^2 \times L}{4CU_2} + V_{C1} = \frac{\Delta I_L^2 \times L}{8CU_2} + V_{C1}$$



V_C maximal ou minimal quand $\frac{dV_C}{dt} = \Delta I_L(t) = 0$

$$V_C(t) = \frac{(U_1 - U_2) \times t^2}{2LC} + \frac{\Delta I_L}{2C} t + V_{C0}$$

$$V_{C,min} = -\frac{{\Delta I_L}^2 \times L}{8C(U_1 - U_2)} - V_{C0}$$

$$V_C(t) = -\frac{U_2 \times t^2}{2LC} + \frac{\Delta I_L}{2C} t + V_{C1}$$

$$V_{C,max} = \frac{\Delta I_L^2 \times L}{8CU_2} + V_{C1}$$

$$\Delta V_C = V_{C,max} - V_{C,min}$$

$$\Delta V_C = \frac{\Delta I_L^2 \times L}{8CU_2} + \frac{\Delta {I_L}^2 \times L}{8C(U_1 - U_2)} = \frac{\Delta I_L^2 \times L}{8C} \frac{U_1}{U_2(U_1 - U_2)}$$

$$\Delta I_L = \frac{(U_1 - U_2)T_1}{L}$$

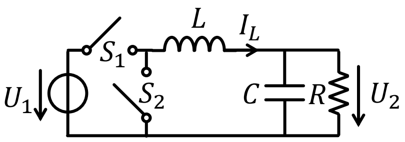
$$V_R = U_1 \times \frac{T_1}{T} \quad \rightarrow T_1 = T\frac{U_2}{U_1} \quad \rightarrow \Delta I_L = \frac{(U_1 - U_2)U_2}{V_1}\frac{T}{L}$$

$$\Delta V_C = \left(\frac{(U_1 - U_2)U_2}{U_1}\frac{T}{L}\right)^2 \frac{L}{8C}\frac{U_1}{U_2(U_1 - U_2)} = \frac{(V_1 - U_2)U_2}{U_1}\frac{T^2}{8LC}$$

$$\boxed{\Delta U_2 = \Delta V_C = \frac{U_2(U_1 - U_2)}{U_1}\frac{T^2}{8LC}}$$

Convertisseur BUCK en conduction continue

BUCK-Wandler in Dauerleitung



C grande / C gross

$\Delta U_2 \ll \bar{U}_2$

$U_2 \approx \bar{U}_2$

Avec des composants (S_1, S_2, L, C) idéals: $\bar{P}_2 = \bar{P}_1$

Mit idealen Komponenten: (S_1, S_2, L, C) $\bar{P}_2 = \bar{P}_1$

$$\boxed{U_2 = U_1 \times \frac{T_1}{T}}$$

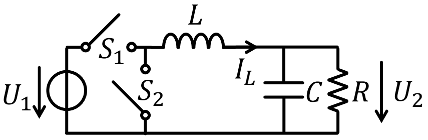
$$\boxed{\bar{I}_L = \bar{I}_R = \frac{U_2}{R}}$$

$$\boxed{\Delta I_L = \frac{(U_1 - U_2)U_2}{U_1}\frac{T}{L}}$$

$$\boxed{I_{L0} = \bar{I}_L - \frac{\Delta I_L}{2}}$$

$$\boxed{\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2}}$$

$$\boxed{\Delta U_2 = \Delta V_C = \frac{U_2(U_1 - U_2)}{U_1}\frac{T^2}{8LC}}$$



$8\,V \leq U_1 \leq 12\,V$

$50\,\Omega \leq R \leq 100\,\Omega$

$U_2 = 5\,V$

Exercice:

La fréquence de commutation d'un convertisseur Buck est $f_T = \frac{1}{T} = 300\,kHz$.

Dimensionnez l'inductance et le condensateur pour:

$$I_{L0} \geq 0\,A$$
$$\Delta U_2 \leq 5\%\,U_2$$

Aufgabe:

Die Schaltfrequenz eines Tiefsetzstellers beträgt $f_T = \frac{1}{T} = 300\,kHz$.

Dimensionieren Sie die Induktivität und den Kondensator für:

$$I_{L0} \geq 0\,A$$
$$\Delta U_2 \leq 5\%\,U_2$$

Convertisseur BUCK en conduction discontinue
BUCK-Wandler in diskontinuierlicher Leitung

Le courent moyenne est donné par la consommation de la charge. Si ce courant devient trop petit, Ça implique un courant minimal plus petit que zéro.

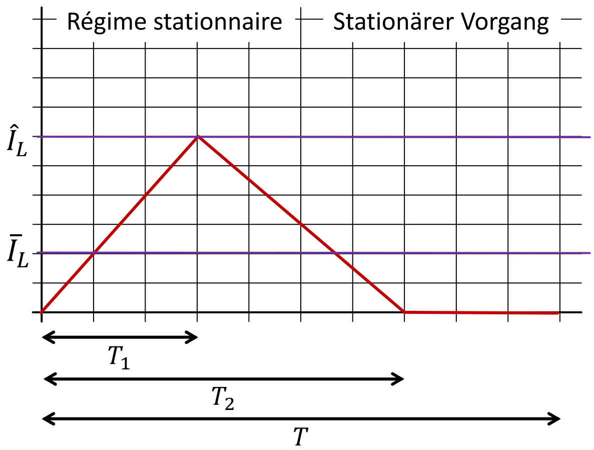
$$I_{L0} < 0$$

Le circuit permet pas ça!
Le circuit rentre dans son mode de conduction discontinue!

Der durchschnittliche Strom ergibt sich aus dem Verbrauch der Last. Wenn dieser Strom zu klein wird, impliziert dies einen minimalen Strom kleiner als Null.

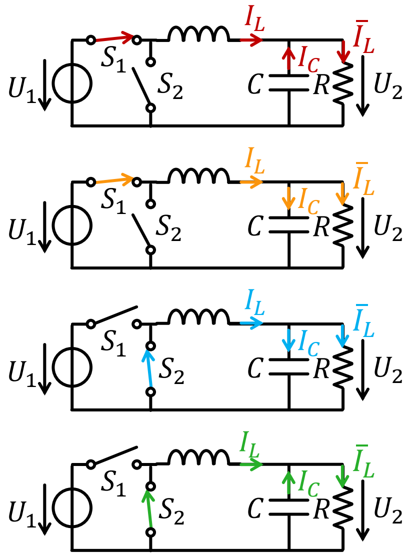
$$I_{L0} < 0$$

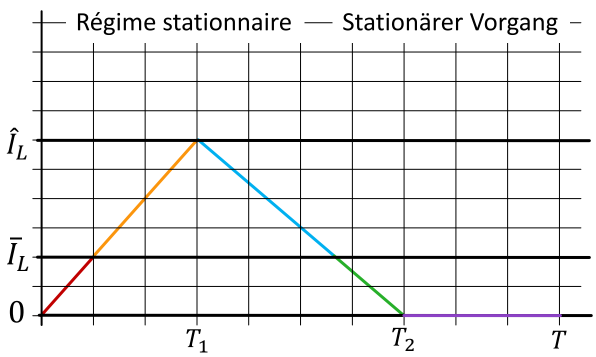
Das lässt die Schaltung nicht zu!
Die Schaltung geht in den diskontinuierlichen Leitungsmodus!

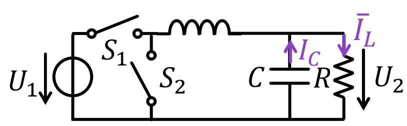


$$\bar{I}_L = \frac{\hat{I}_L T_2}{2T}$$

Courants en conduction discontinue
Ströme in diskontinuierlicher Leitung







Condensateur / Kondensator

$$V_C(t) = \frac{1}{C}\int_{t_0}^{t} I_C(\tau)d\tau + V_C(t_0)$$

$t = 0 \rightarrow T_1$

$I_C(t) = I_L(t) - \bar{I}_L = \tilde{I}_L(t)$

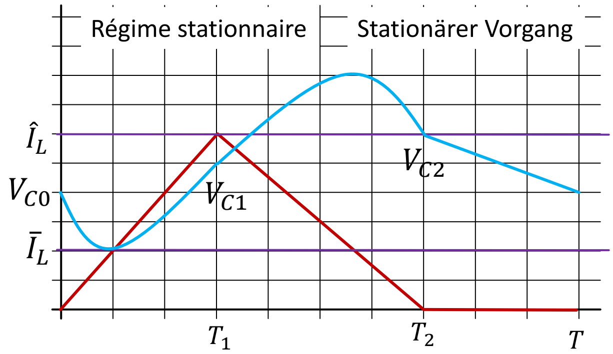
$\tilde{I}_L(t) = \hat{I}_L t - \bar{I}_L$

$$V_C(t) = \frac{\hat{I}_L t^2}{2C} - \frac{\bar{I}_L}{C}t + V_{C0}$$

$t = T_1 \rightarrow T_2$

$\tilde{I}_L(t) = \hat{I}_L - \bar{I}_L - \hat{I}_L t$

$$V_C(t) = -\frac{\hat{I}_L t^2}{2C} + \frac{\hat{I}_L - \bar{I}_L}{C}t + V_{C1}$$



$t = T_2 \rightarrow T$

$\tilde{I}_L(t) = -\bar{I}_L$

$$V_C(t) = -\frac{\bar{I}_L}{C}t + V_{C2}$$

$$\Delta V_C = V_{C,max} - V_{C,min}$$

Condensateur / Kondensator

Hes·so Valais Wallis
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

$t = 0 \rightarrow T_1$

$$V_C(t) = \frac{\hat{I}_L t^2}{2C} - \frac{\bar{I}_L}{C}t + V_{C0}$$

$t = T_1 \rightarrow T_2$

$$V_C(t) = -\frac{\hat{I}_L t^2}{2C} + \frac{\hat{I}_L - \bar{I}_L}{C}t + V_{C1}$$

Min when

$$\hat{I}_L t - \bar{I}_L = 0 \rightarrow t = \frac{\bar{I}_L}{\hat{I}_L}$$

$$V_{Cmin} = \frac{\hat{I}_L}{2C}\left(\frac{\bar{I}_L}{\hat{I}_L}\right)^2 - \frac{\bar{I}_L}{C}\left(\frac{\bar{I}_L}{\hat{I}_L}\right) + V_{C0}$$

$$V_{Cmin} = \frac{\bar{I}_L^2}{2C\hat{I}_L} - \frac{\bar{I}_L^2}{C\hat{I}_L} + V_{C0}$$

$$= V_{C0} - \frac{\bar{I}_L^2}{2C\hat{I}_L}$$

Max when

$$\hat{I}_L - \bar{I}_L - \hat{I}_L t = 0 \rightarrow t = \frac{\hat{I}_L - \bar{I}_L}{\hat{I}_L}$$

$$V_{Cmax} = -\frac{\hat{I}_L}{2C}\left(\frac{\hat{I}_L - \bar{I}_L}{\hat{I}_L}\right)^2 + \frac{\hat{I}_L - \bar{I}_L}{C}\left(\frac{\hat{I}_L - \bar{I}_L}{\hat{I}_L}\right) + V_{C1}$$

$$V_{Cmax} = \frac{\left(\hat{I}_L - \bar{I}_L\right)^2}{2C\hat{I}_L} + V_{C1}$$

$$\Delta V_C = \frac{\left(\hat{I}_L - \bar{I}_L\right)^2}{2C\hat{I}_L} + \frac{\bar{I}_L^2}{2C\hat{I}_L} + V_{C1} - V_{C0}$$

Condensateur / Kondensator

$$\boxed{\begin{array}{l} t = 0 \rightarrow T_1 \\ V_C(t) = \dfrac{\hat{I}_L t^2}{2C} - \dfrac{\bar{I}_L}{C} t + V_{C0} \end{array}}$$

$$V_{C1} = \frac{\hat{I}_L T_1^2}{2C} - \frac{\bar{I}_L}{C} T_1 + V_{C0}$$

$$V_{C1} - V_{C0} = \frac{\hat{I}_L T_1^2}{2C} - \frac{\bar{I}_L}{C} T_1$$

$$\Delta V_C = \frac{\left(\hat{I}_L - \bar{I}_L\right)^2}{2C\hat{I}_L} + \frac{\bar{I}_L^2}{2C\hat{I}_L} + \frac{\hat{I}_L T_1^2}{2C} - \frac{\bar{I}_L}{C} T_1$$

Conduction continue ou discontinue?
Kontinuierliche oder diskontinuierliche Leitung?

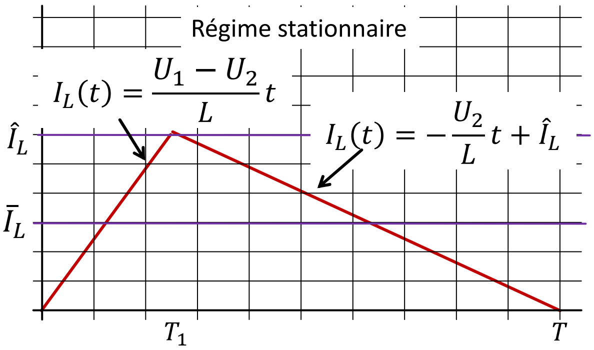
Cas marginal: Le courent dans l'inductance est zéro à $t = 0$, et $t = T$

$$\hat{I}_L = \frac{U_1 - U_2}{L} T_1$$

$$\bar{I}_L = \frac{\hat{I}_L}{2} \qquad \bar{I}_L = \frac{U_2}{R}$$

$$\frac{U_1 - U_2}{2L} T_1 = \frac{U_2}{R} \qquad T_1 = \frac{T U_2}{U_1}$$

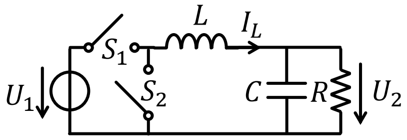
$$\frac{U_1 - U_2}{2L} \frac{T U_2}{U_1} = \frac{U_2}{R} \qquad R = \frac{2LU_1}{T(U_1 - U_2)}$$



$R \leq \dfrac{2LU_1}{T(U_1 - U_2)} \rightarrow$ Conduction continue / Kontinuierliche Leitung

$R > \dfrac{2LU_1}{T(U_1 - U_2)} \rightarrow$ Conduction discontinue / Diskontinuierliche Leitung

Convertisseur BUCK en conduction discontinue
BUCK-Wandler in diskontinuierlicher Leitung



C grande / C gross
$\Delta U_2 \ll \bar{U}_2$
$U_2 \approx \bar{U}_2$

$$\boxed{\begin{array}{c}\text{Avec des composants}\\ (S_1, S_2, L, C)\ \text{idéals:}\\ \bar{P}_2 = \bar{P}_1\end{array}}$$

$$\boxed{\begin{array}{c}\text{Mit idealen Komponenten:}\\ (S_1, S_2, L, C)\\ \bar{P}_2 = \bar{P}_1\end{array}}$$

Donnés: U_1, T_1, R, L, T

$$\boxed{T_2 = \frac{T_1}{2} + \sqrt{\frac{T_1^2}{4} + \frac{2TL}{R}}}$$

$$\boxed{U_2 = U_1 \frac{T_1}{T_2}}$$

Donnés: U_1, U_2, R, L, T

$$\boxed{T_1 = \sqrt{\frac{2TL}{R} \frac{U_2^2}{U_1(U_1 - U_2)}}}$$

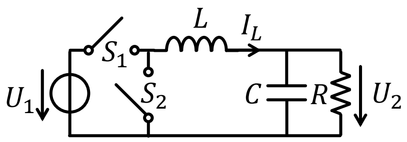
$$\boxed{T_2 = \frac{U_1}{U_2} T_1}$$

$$\bar{I}_L = \frac{\hat{I}_L T_2}{2T} \qquad \bar{I}_L = \frac{U_2}{R}$$

$$\hat{I}_L = \frac{U_1 - U_2}{L} T_1 \qquad \hat{I}_L = \frac{U_2}{L}(T_2 - T_1)$$

$$\boxed{\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_L)^2}{2C\hat{I}_L} + \frac{\bar{I}_L^2}{2C\hat{I}_L} + \frac{\hat{I}_L T_1^2}{2C} - \frac{\bar{I}_L}{C} T_1}$$

Convertisseur BUCK en conduction continue
BUCK-Wandler in Dauerleitung



C grande / C gross
$\Delta U_2 \ll \bar{U}_2$
$U_2 \approx \bar{U}_2$

$$\boxed{\begin{array}{c}\text{Avec des composants}\\ (S_1, S_2, L, C)\ \text{idéals:}\\ \bar{P}_2 = \bar{P}_1\end{array}}$$

$$\boxed{\begin{array}{c}\text{Mit idealen Komponenten:}\\ (S_1, S_2, L, C)\\ \bar{P}_2 = \bar{P}_1\end{array}}$$

$$\boxed{U_2 = U_1 \times \frac{T_1}{T}}$$

$$\boxed{\bar{I}_L = \bar{I}_R = \frac{U_2}{R}}$$

$$\boxed{\Delta I_L = \frac{(U_1 - U_2)U_2}{U_1} \frac{T}{L}}$$

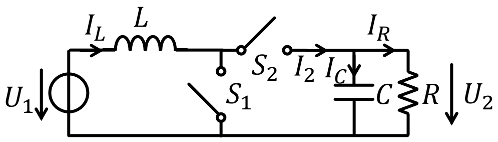
$$\boxed{I_{L0} = \bar{I}_L - \frac{\Delta I_L}{2}}$$

$$\boxed{\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2}}$$

$$\boxed{\Delta U_2 = \Delta V_C = \frac{U_2(U_1 - U_2)}{U_1} \frac{T^2}{8LC}}$$

Convertisseur BOOST - Boost-Wandler

Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

$0 \leq t \leq T_1$	$V_L =$
$I_L(t) =$	

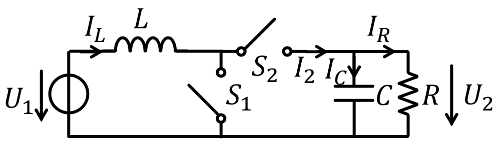
$T_1 \leq t \leq T$	$V_L =$
$I_L(t) =$	

- À temps $t = 0$ et $t = T$ l'interrupteur S_1 est fermé et l'interrupteur S_2 est ouvert.
- À $t = T_1$ l'interrupteur S_1 est ouvert et l'interrupteur S_2 est fermé.
- Et ainsi de suite
- Régime stationnaire: $I_L(T) = I_L(0)$
- Dessinez I_L, I_2 et I_R
- Trouvez $U_2, \hat{I}_L, \bar{I}_L$ et I_{L0}, Données: U_1, T_1, T, L, R, C

- Bei $t = 0$ und $t = T$ ist Schalter S_1 geschlossen und Schalter S_2 geöffnet.
- Bei $t = T_1$ ist Schalter S_1 geöffnet und Schalter S_2 geschlossen.
- Und so weiter
- Stationärer Vorgang : $I_L(T) = I_L(0)$
- Zeichnen Sie I_L, I_2 und I_R
- Finden Sie $U_2, \hat{I}_L, \bar{I}_L$ und I_{L0},
- Gegeben : U_1, T_1, T, L, R, C

Convertisseur BOOST - Boost-Wandler

Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



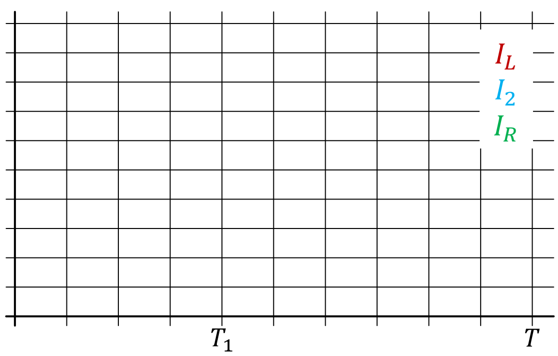
$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

$0 \leq t \leq T_1$	$V_L =$
$I_L(t) =$	

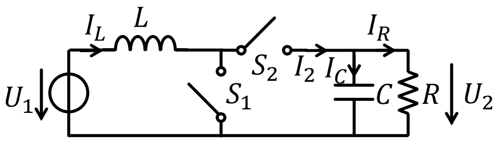
$T_1 \leq t \leq T$	$V_L =$
$I_L(t) =$	

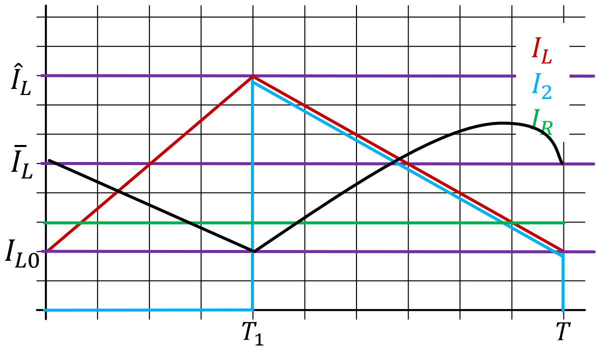
- Trouvez $U_2, \hat{I}_L, \bar{I}_L$ et I_{L0},
- Données: U_1, T_1, T, L, R, C

- Finden Sie $U_2, \hat{I}_L, \bar{I}_L$ und I_{L0},
- Gegeben : U_1, T_1, T, L, R, C



Convertisseur BOOST - Boost-Wandler





$$\bar{I}_2 = I_R = \frac{U_2}{R} \qquad I_2 = \bar{I}_2 + \Delta I_2$$

$$\boxed{\hat{I}_L = \frac{U_2^2}{RU_1} + \frac{U_1 T_1}{2L}}$$

$$V_C(t) = \frac{1}{C}\int_{t_0}^{t} I_C(\tau)d\tau + V_C(t_0)$$

$$0 \leq t \leq T_1$$

$$\Delta I_2(t) = -\bar{I}_2$$

$$V_C(t) = -\frac{\bar{I}_2 t}{C} + V_{C0}$$

$$T_1 \leq t \leq T$$

$$\Delta I_2(t) = -\Delta I_L \frac{t}{T - T_1} + \hat{I}_L - \bar{I}_2$$

$$V_C(t) = -\frac{\Delta I_L}{2C(T - T_1)}t^2 + \frac{(\hat{I}_L - \bar{I}_2)}{C}t + V_{C1}$$

Hes·so// VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

$$V_C(t) = -\bar{I}_2 t + V_{C0}$$

$$V_{Cmin} = V_C(T_1) = V_{C1}$$

$$\Delta I_2(t) = -\Delta I_L \frac{t}{T - T_1} + \hat{I}_L - \bar{I}_2$$

$$t = \frac{\hat{I}_L - \bar{I}_2}{\Delta I_L}(T - T_1)$$

$$V_C(t) = -\frac{\Delta I_L}{2C(T - T_1)}t^2 + \frac{(\hat{I}_L - \bar{I}_2)}{C}t + V_{C1}$$

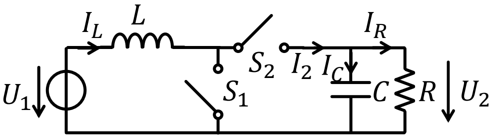
$$V_{Cmax} = -\frac{\Delta I_L}{2C(T - T_1)}\left((T - T_1)\frac{\hat{I}_L - \bar{I}_2}{\Delta I_L}\right)^2 + (\hat{I}_L - \bar{I}_2)\frac{\hat{I}_L - \bar{I}_2}{\Delta I_L C}(T - T_1) + V_{C1}$$

$$\Delta V_C = -\frac{(\hat{I}_L - \bar{I}_2)^2 (T - T_1)}{2\Delta I_L C} + \frac{(\hat{I}_L - \bar{I}_2)^2 (T - T_1)}{\Delta I_L C} = \frac{(\hat{I}_L - \bar{I}_2)^2 (T - T_1)}{2\Delta I_L C}$$

$$\boxed{T_1 = T\frac{U_2 - U_1}{U_2}} \qquad \boxed{T - T_1 = T\frac{U_1}{U_2}} \qquad \boxed{\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_2)^2 T U_1}{2\Delta I_L C U_2}}$$

Convertisseur BOOST en conduction continue
BOOST-Wandler in Dauerleitung

Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



$$U_2 = U_1 \frac{T}{T - T_1}$$

$$T_1 = T \frac{U_2 - U_1}{U_2}$$

$$\Delta I_L = \frac{U_1(U_2 - U_1)}{U_2}\frac{T}{L}$$

Pour un circuit ou C est grande.
Für eine Schaltung, bei der C gross ist.

$$\Delta U_2 \ll \bar{U}_2$$
$$U_2 \approx \bar{U}_2$$

$$\bar{I}_L = \frac{U_2^2}{RU_1}$$

$$\bar{I}_2 = \frac{U_2}{R}$$

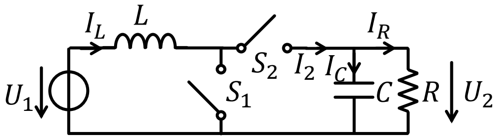
$$\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2} = \frac{U_2^2}{RU_1} + \frac{T}{2L}\frac{U_1(U_2 - U_1)}{U_2}$$

$$\Delta U_2 = \Delta V_C = \frac{\left(\hat{I}_L - \bar{I}_2\right)^2 T U_1}{2\Delta I_L C U_2}$$

$$I_{L0} = \bar{I}_L - \frac{\Delta I_L}{2} = \frac{U_2^2}{RU_1} - \frac{T}{2L}\frac{U_1(U_2 - U_1)}{U_2}$$

Convertisseur BOOST - Boost-Wandler

Hes·so VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



- À temps $t = 0$ et $t = T$ l'interrupteur S_1 est fermé et l'interrupteur S_2 est ouvert.
- À $t = T_1$ l'interrupteur S_1 est ouvert et l'interrupteur S_2 est fermé.
- À $t = T_2$ le courent est $0\ A$
- Régime stationnaire: $I_L(T) = I_L(0) = 0\ A$
- Dessinez I_L, I_2 et I_R
- Trouvez $T_1, T_2, \hat{I}_L$, et $\bar{I}_L$, Données: U_1, U_2, T, L, R, C

$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

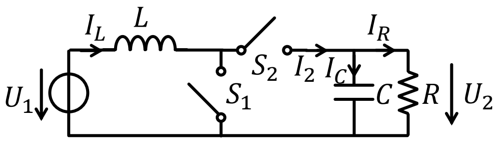
$0 \leq t \leq T_1$	$V_L =$
$I_L(t) =$	

$T_1 \leq t \leq T_2$	$V_L =$
$I_L(t) =$	

- Bei $t = 0$ und $t = T$ ist Schalter S_1 geschlossen und Schalter S_2 geöffnet.
- Bei t $= T_1$ ist Schalter S_1 geöffnet und Schalter S_2 geschlossen.
- Bei t $= T_1$ ist der Strom $0A$.
- Stationärer Vorgang : $I_L(T) = I_L(0) = 0\ A$
- Zeichnen Sie I_L, I_2 und I_R
- Finden Sie $T_1, T_2, \hat{I}_L$, und $\bar{I}_L$
- Gegeben :U_1, U_2, T, L, R, C

Convertisseur BOOST - Boost-Wandler

Hes·so// VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften



- Dessinez I_L, I_2 et I_R
- Trouvez $T_1, T_2, \hat{I}_L$, et $\bar{I}_L$, Données: U_1, U_2, T, L, R, C
- Finden Sie $T_1, T_2, \hat{I}_L$, und $\bar{I}_L$
- Gegeben :U_1, U_2, T, L, R, C

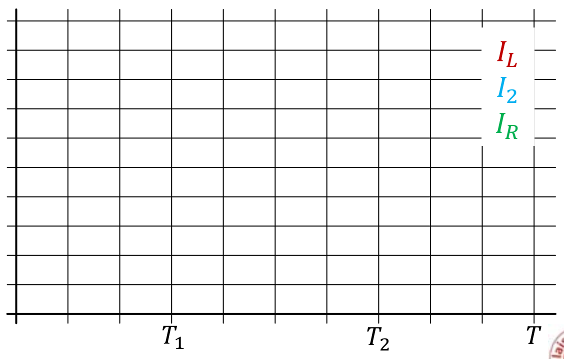
$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

$0 \le t \le T_1 \qquad V_L =$

$I_L(t) =$

$T_1 \le t \le T_2 \qquad V_L =$

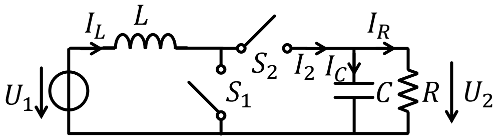
$I_L(t) =$



Convertisseur BOOST - Boost-Wandler

Hes·so// VALAIS WALLIS
Haute Ecole d'Ingénierie
Hochschule für Ingenieurwissenschaften

Trouvez/Finden Sie $T_1, T_2, \hat{I}_L$ et $\bar{I}_L$



$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau + I_L(t_0)$$

$0 \le t \le T_1 \qquad V_L = U_1$

$$I_L(t) = \frac{U_1 t}{L}$$

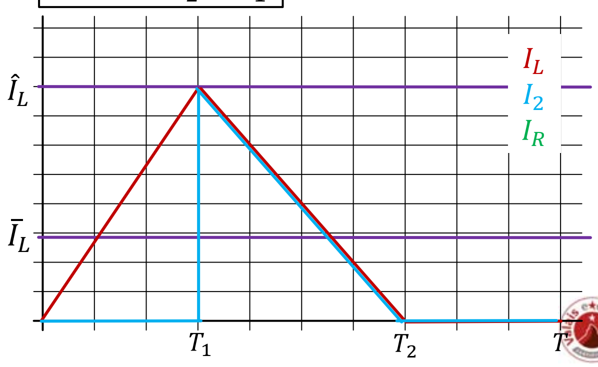
$T_1 \le t \le T_2 \qquad V_L = U_1 - U_2$

$$I_L(t) = \frac{(U_1 - U_2)t}{L} + \hat{I}_L$$

$$\hat{I}_L = \frac{U_1 T_1}{L} \qquad I_{L0} = \frac{(U_1 - U_2)(T_2 - T_1)}{L} + \hat{I}_L$$

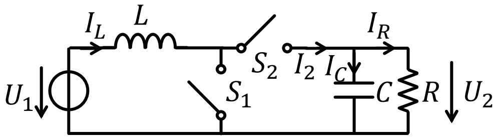
$$\frac{(U_1 - U_2)(T_2 - T_1)}{\cancel{L}} + \frac{U_1 T_1}{\cancel{L}} = 0$$

$$\boxed{U_2 = U_1 \frac{T_2}{T_2 - T_1}}$$



Convertisseur BOOST - Boost-Wandler

Trouvez/Finden Sie $T_1, T_2, \hat{I}_L$ et $\bar{I}_L$



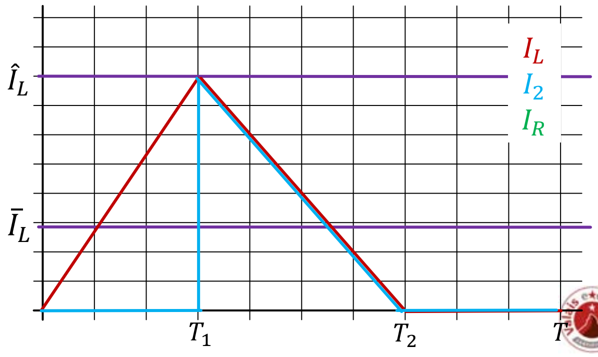
$$\boxed{U_2 = U_1 \frac{T_2}{T_2 - T_1}} \qquad \boxed{\hat{I}_L = \frac{U_1 T_1}{L}}$$

$$\bar{I}_L = \frac{\hat{I}_L T_2}{2T} = \frac{U_1 T_1 T_2}{2LT}$$

$$\bar{I}_2 = \frac{U_2}{R}$$

$$\bar{I}_2 = \frac{\hat{I}_L (T_2 - T_1)}{2T} = \frac{U_1 T_1 (T_2 - T_1)}{2TL}$$

$$\frac{U_2}{R} = \frac{U_1 T_1 (T_2 - T_1)}{2TL}$$



$$\frac{U_2}{R} = \frac{U_1 T_1 (T_2 - T_1)}{2TL}$$

$$\frac{U_2}{R} = \frac{U_1 T_1 \left(\frac{U_2}{U_2 - U_1} T_1 - T_1 \right)}{2TL}$$

$$\frac{U_2}{R} = \frac{\frac{U_1^2}{U_2 - U_1} T_1^2}{2TL}$$

$$T_1^2 = \frac{2LTU_2 (U_2 - U_1)}{RU_1^2}$$

$$\boxed{T_1 = \sqrt{\frac{2LTU_2 (U_2 - U_1)}{RU_1^2}}}$$

$$U_2 = U_1 \frac{T_2}{T_2 - T_1}$$

$$\boxed{T_2 = \frac{U_2}{U_2 - U_1} T_1}$$

$$\boxed{\hat{I}_L = \frac{U_1 T_1}{L}}$$

$$\boxed{\bar{I}_L = \frac{U_1 T_1 T_2}{2LT}}$$

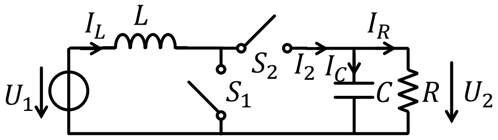
$$\frac{U_2}{R} = \frac{U_1 T_1 (T - T_1)}{2TL}$$

$$2TLU_2 = RU_1 \frac{(U_2 - U_1)}{U_2} T \left(T - \frac{(U_2 - U_1)}{U_2} T \right)$$

$$2LU_2 = R \frac{TU_1^2 (U_2 - U_1)}{U_2^2}$$

$$R \geq \frac{2LU_2^3}{TU_1^2 (U_2 - U_1)}$$

Convertisseur BOOST - Boost-Wandler



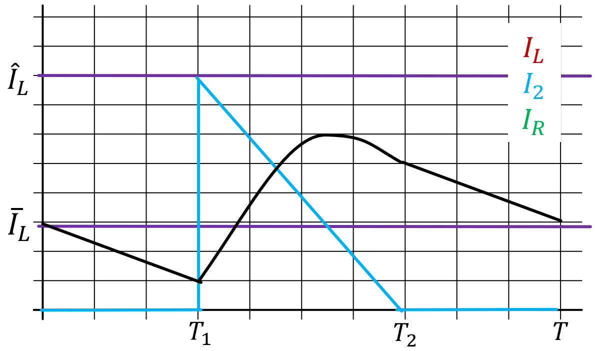
$$\tilde{I}_2 = \hat{I}_L - \hat{I}_L \frac{t}{T_2 - T_1} - \bar{I}_2$$

$$t = \frac{\hat{I}_L - \bar{I}_2}{\hat{I}_L}(T_2 - T_1)$$

$$V_C = \frac{1}{C}(\hat{I}_L - \bar{I}_2)t - \frac{\hat{I}_L}{2C}\frac{t^2}{T_2 - T_1} + V_{C1}$$

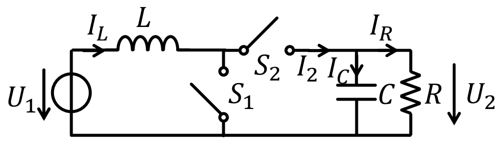
$$V_{C,max} = \frac{(\hat{I}_L - \bar{I}_2)^2}{2\hat{I}_L C}(T_2 - T_1) + V_{C1}$$

$$\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_2)^2}{2\hat{I}_L C}(T_2 - T_1)$$



$$\tilde{I}_L = -\bar{I}_2$$

Convertisseur BOOST en conduction discontinue
BOOST-Wandler in diskontinuierlicher Leitung



$$\boxed{T_1 = \sqrt{\frac{2LTU_2(U_2 - U_1)}{RU_1^2}}}$$

$$\boxed{U_2 = U_1 \frac{T_2}{T_2 - T_1}} \qquad \boxed{T_2 = \frac{U_2}{U_2 - U_1}T_1}$$

Pour un circuit ou C est grande.

Für eine Schaltung, bei der C gross ist.

$$\Delta U_2 \ll \bar{U}_2$$
$$U_2 \approx \bar{U}_2$$

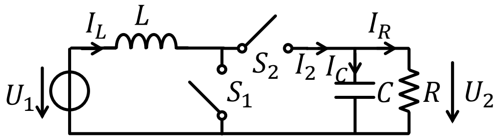
$$\boxed{\Delta I_L = \frac{U_1}{L}T_1}$$

$$\boxed{R \geq \frac{2LU_2^3}{TU_1^2(U_2 - U_1)}}$$

$$\boxed{\bar{I}_L = \frac{\hat{I}_L(T_2 - T_1)}{2T} = \frac{U_1 T_1 (T_2 - T_1)}{2TL}}$$

$$\boxed{\bar{I}_2 = \frac{U_2}{R}}$$

$$\boxed{\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_2)^2 T U_1}{2\Delta I_L C U_2}}$$



$3\,V \le U_1 \le 4.5\,V$

$500\,\Omega \le R \le 2000\,\Omega$

$U_2 = 9\,V$

Exercice:

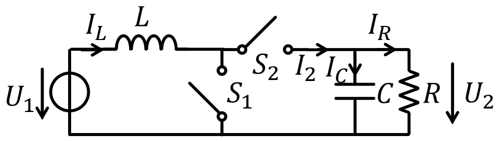
La fréquence de commutation d'un convertisseur Boost est $f_T = \frac{1}{T} = 250\ kHz$.

Dimensionnez l'inductance et le condensateur pour:

- Convertisseur en conduction continue
- $\Delta I_L \le 30\%\ \bar{I}_L$
- $\Delta U_2 \le 5\%\ U_2$

Aufgabe:

- Die Schaltfrequenz eines Boost-Wandlers ist $f_T = \frac{1}{T} = 250\ kHz$.
- Dimensionieren Sie die Induktivität und den Kondensator für:
- Wandler im kontinuierliche Leitung
- $\Delta I_L \le 30\%\ \bar{I}_L$
- $\Delta U_2 \le 5\%\ U_2$



$$\boxed{\Delta I_L = \frac{U_1(U_2 - U_1)}{U_2}\frac{T}{L}} \qquad \boxed{\bar{I}_L = \frac{U_2^2}{RU_1}}$$

$3\,V \le U_1 \le 4.5\,V$

$500\,\Omega \le R \le 2000\,\Omega$

$U_2 = 9\,V$

$$\boxed{\Delta U_2 = \Delta V_C = \frac{\left(\hat{I}_L - \bar{I}_2\right)^2 T U_1}{2\Delta I_L C U_2}}$$

$$\frac{\Delta I_L}{\bar{I}_L} = \frac{U_1(U_2 - U_1)}{U_2}\frac{T}{L}\frac{RU_1}{U_2^2} = \frac{U_1^2(U_2 - U_1)}{U_2^3}\frac{TR}{L}$$

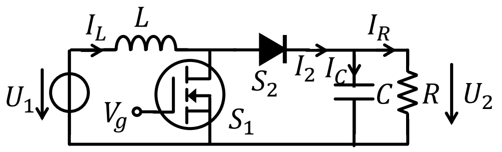
$$\frac{\Delta U_2}{U_2} = \frac{\left(\hat{I}_L - \bar{I}_2\right)^2 T U_1}{2\Delta I_L C U_2^2} \le 0.05$$

$$L > \frac{U_1^2(U_2 - U_1)}{U_2^3} TR \frac{\bar{I}_L}{\Delta I_L}$$

$$C \ge \frac{\left(\hat{I}_L - \bar{I}_2\right)^2 T U_1}{2\Delta I_L U_2^2 \times 0.05}$$

$$\boxed{\bar{I}_2 = \frac{U_2}{R}}$$

$$\boxed{\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2} = \frac{U_2^2}{RU_1} + \frac{T}{2L}\frac{U_1(U_2 - U_1)}{U_2}}$$



$U_1 = 4.5\ V$ $\qquad U_2 = 9\ V$

$R = 2\ \mathrm{k\Omega}$ $\qquad L = 3.3\ mH, R_s = 10\ \Omega$

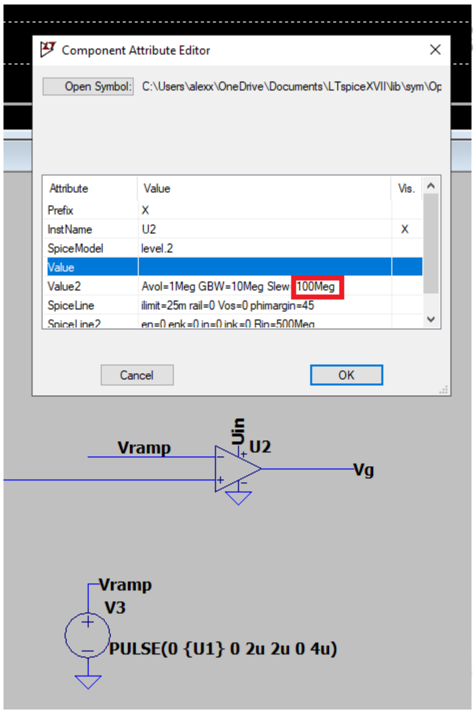
$C = 1\ \mu F$ $\qquad f_T = \frac{1}{T} = 250\ kHz$

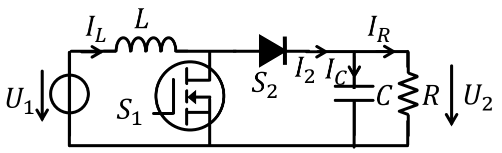
N-MOS : Si7368DP

Diode : 1N914

$V_{ref} = 2.25\ V$

$$\boxed{U_2 = U_1 \frac{T}{T - T_1}}$$





$U_1 = 4.0\ V$ $\qquad U_2 = 9\ V$

$R = 1000\ \Omega$ $\qquad L = 3.3\ mH$

Exercice:

La fréquence de commutation d'un convertisseur Boost est $f_T = \frac{1}{T} = 250\ kHz$.

S_1 est implémenté avec un MOSFET avec $V_{DS,on} = 0.2$ V

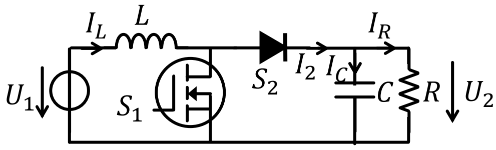
S_2 est implémenté avec une diode avec $V_F = 0.7\ V$

Calculez le temps de conduction T_1

Calculez la perte et le rendement

Aufgabe:

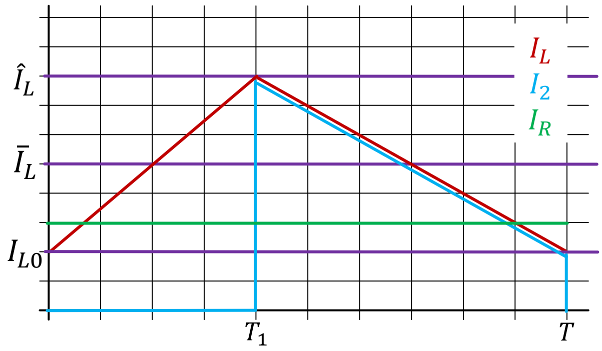
- Die Schaltfrequenz eines Boost-Wandlers ist $f_T = \frac{1}{T} = 250\ kHz$.
- S_1 ist mit einem MOSFET mit $V_{DS,on} = 0.2$ V Implementiert
- S_2 ist mit einer Diode mit $V_F = 0.7\ V$ Implementiert
- Berechnen Sie die Leitungszeit T_1
- Berechnen Sie den Verlust und den Wirkungsgrad

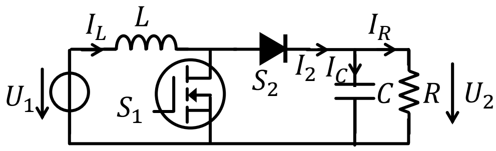


$U_1 = 4.0\ V \qquad U_2 = 9\ V$

$R = 1000\ \Omega \qquad L = 3.3\ mH$

$$R_{crit} = \frac{2LU_2^3}{TU_1^2(U_2 - U_1)} = 15\ k\Omega$$





$U_1 = 4.0\ V \qquad U_2 = 9\ V$

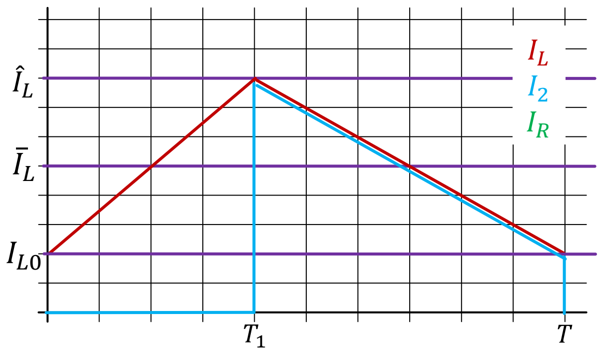
$R = 1000\ \Omega \qquad L = 3.3\ mH$

$$R_{crit} = \frac{2LU_2^3}{TU_1^2(U_2 - U_1)} = 15\ k\Omega$$

$$I_L(t) = \frac{1}{L}\int_{t_0}^{t} V_L(\tau)d\tau$$

$$\Delta I_L = \frac{1}{L}(U_1 - V_{DS})T_1$$

$$\Delta I_L = \frac{1}{L}(U_1 - V_F - U_2)(T - T_1)$$



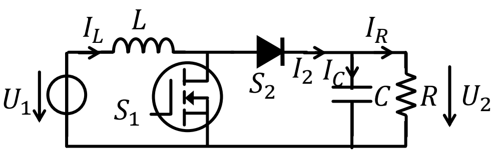
$$(U_1 - V_{DS})T_1 = -(U_1 - V_F - U_2)(T - T_1)$$

$$(U_1 - V_{DS})T_1 = -(U_1 - V_F - U_2)(T - T_1)$$

$$(U_1 - V_{DS} - U_1 + V_F + U_2)T_1 = -(U_1 - V_F - U_2)T$$

$$T_1 = \frac{(U_2 + V_F - U_1)}{(U_2 + V_F - V_{DS})}T = \frac{(9 + 0.7 - 4)}{(9 + 0.7 - 0.2)}4\mu s = 2.4\ \mu s$$

$$\boxed{T_1 = T\frac{U_2 - U_1}{U_2} = 2.22\ \mu s}$$



$$\bar{P}_1 = U_1\bar{I}_L$$

$$\eta = \frac{\bar{P}_2}{\bar{P}_1} = \frac{U_2}{U_1}\frac{(T - T_1)}{T} = \frac{9}{4} \times \frac{4\mu s - 2.4\ \mu s}{4\mu s} = 0.9 = 90\%$$

$$\bar{P}_2 = U_2\bar{I}_2 = \frac{U_2^2}{R}$$

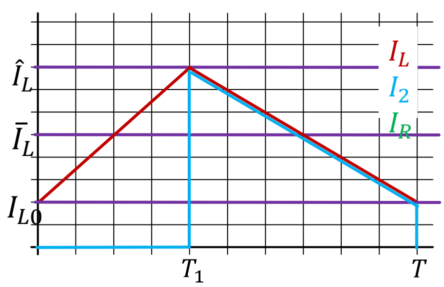
$$\bar{P}_2 = \frac{U_2^2}{R} = 81\ mW$$

$$\bar{I}_2 = \frac{\bar{I}_L(T - T_1)}{T}$$

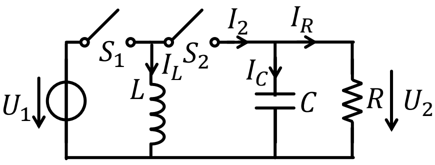
$$\bar{P}_1 = \frac{\bar{P}_2}{\eta} = 90\ mW$$

$$\bar{P}_2 = U_2\frac{\bar{I}_L(T - T_1)}{T}$$

$$\bar{P}_{Loss} = 9\ mW$$



Convertisseur BUCK-BOOST



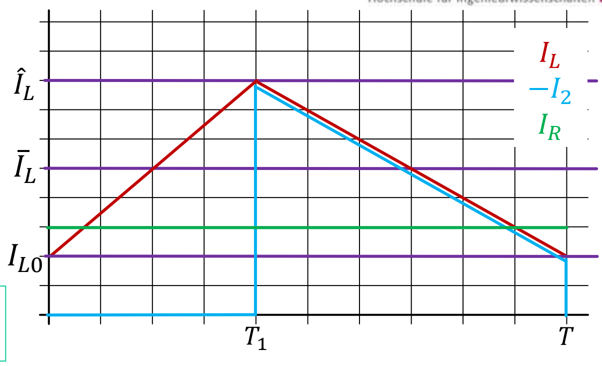
$\Delta I_L =$

$U_2 =$ $T_1 =$

$\hat{I}_L =$

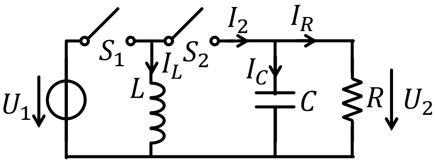
$\bar{I}_L =$

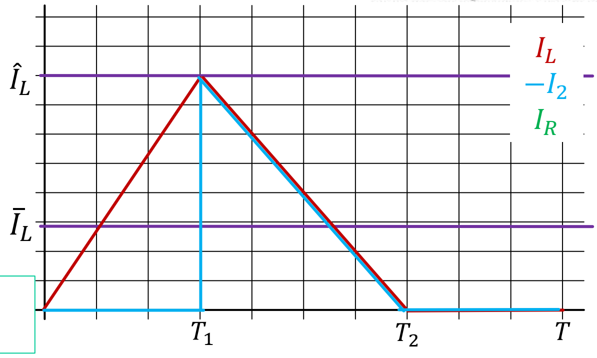
$\bar{I}_2 =$



$$\Delta U_2 = \frac{(\hat{I}_L - \bar{I}_2)^2 (T - T_1)}{2\Delta I_L C} = \frac{(\hat{I}_L - \bar{I}_2)^2}{2\Delta I_L C} \frac{U_1}{U_1 - U_2} T$$

Convertisseur BUCK-BOOST





$\Delta I_L =$

$U_2 =$ $\qquad T_2 =$

$\hat{I}_L =$

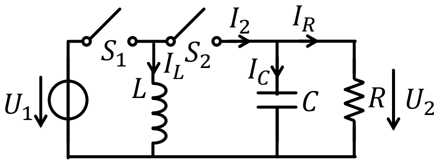
$\bar{I}_L =$

$\bar{I}_2 =$

$$\Delta U_2 = \Delta V_C = \frac{(\hat{I}_L - \bar{I}_2)^2}{2\hat{I}_L C}(T_2 - T_1)$$

$T_1 =$ $\qquad R \geq$

Convertisseur BUCK-BOOST en conduction continue



Pour un circuit ou C est grande.
Ça va dire:

$$\Delta U_2 \ll \bar{U}_2$$
$$U_2 \approx \bar{U}_2$$

$$\boxed{\bar{I}_2 = \frac{U_2}{R}}$$

$$\boxed{\Delta U_2 = \frac{(\hat{I}_L - \bar{I}_2)^2}{2\Delta I_L C}\frac{U_1}{U_1 - U_2}T}$$

$$\boxed{U_2 = -U_1\frac{T_1}{T - T_1}} \quad \boxed{T_1 = T\frac{U_2}{U_2 - U_1}}$$

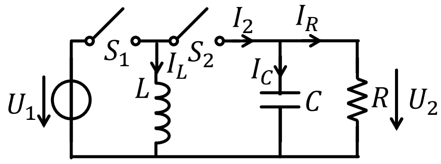
$$\boxed{\Delta I_L = \frac{U_1 T_1}{L} = \frac{U_2(T - T_1)}{L}}$$

$$\boxed{\bar{I}_L = -\frac{U_2}{R}\frac{T}{T - T_1}}$$

$$\boxed{\hat{I}_L = \bar{I}_L + \frac{\Delta I_L}{2}}$$

$$\boxed{I_{L0} = \bar{I}_L - \frac{\Delta I_L}{2}}$$

Convertisseur BUCK-BOOST en conduction discontinue



$$T_1 = -\frac{U_2}{U_1}\sqrt{\frac{2LT}{R}}$$

$$U_2 = -T_1 U_1 \sqrt{\frac{R}{2LT}}$$

$$T_2 = T_1 \frac{U_2 - U_1}{U_2}$$

$$R \geq \frac{2L}{T}\frac{(U_2 - U_1)^2}{U_1^2}$$

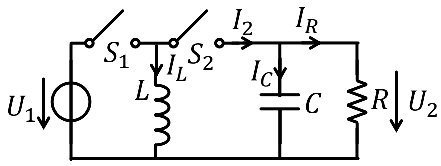
$$\Delta I_L = \frac{U_1 T_1}{L} = \frac{U_2(T_2 - T_1)}{L}$$

$$\Delta U_2 = \Delta V_C = \frac{\left(\hat{I}_L - \bar{I}_2\right)^2}{2\hat{I}_L C}(T_2 - T_1)$$

$$\hat{I}_L = \Delta I_L$$

$$\bar{I}_L = \frac{\Delta I_L}{2}\frac{T_2}{T} = -\frac{U_2}{R}\frac{T_2}{T_2 - T_1}$$

$$\bar{I}_2 = \frac{U_2}{R}$$



$$3\,V \leq U_1 \leq 8\,V$$

$$200\,\Omega \leq R \leq 1\,k\Omega$$

$$U_2 = -5\,V$$

Exercice:

La fréquence de commutation d'un convertisseur Buck-Boost est $f_T = \frac{1}{T} = 50\,kHz$.

Dimensionnez l'inductance et le condensateur pour:

Pour $200\,\Omega \leq R \leq 1k\Omega,\ I_{L0} \geq 0\,A$

$\Delta U_2 \leq 5\%\,U_2$

Trouvez L et C minimal

Calculez T_1 pour $U_1 = 3\,V$ et $U_1 = 8V$